

AKHALTSIKHE, Georgia

WMO: 375140

Lat: 41.6506N

Lon: 43.0069E

Elev: 994

StdP: 89.94

Time Zone: 4.00 (E04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-11.2	-16.2	1.0	-13.1	-13.8	1.3	-10.4	8.4	5.0	7.2	3.0	0.0	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.1	33.0	21.6	31.1	20.7	29.3	19.9	22.2	31.2	21.3	29.7	20.4	28.1	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	15.9	26.4	18.5	15.0	25.3	17.7	14.3	24.3	70.4	31.7	66.8	29.8	63.5	28.2	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.0	5.5	-16.3	36.2	3.6	1.5	-18.9	37.3	-21.0	38.2	-23.1	39.0	-25.7	40.1		
(5)			-16.5	23.6	3.6	0.9	-19.1	24.3	-21.2	24.8	-23.2	25.3	-25.8	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-1.6	0.6	5.3	9.8	14.9	19.1	21.9	22.4	17.6	11.7	4.8	0.3
	DBStd	8.97	4.42	4.41	3.52	3.60	2.87	2.52	2.54	2.83	3.34	3.06	3.43	3.99
	HDD10.0	1302	360	263	151	48	2	0	0	0	1	19	157	301
	HDD18.3	3140	618	497	405	258	112	21	4	4	52	206	405	559
	CDD10.0	1524	0	0	4	41	153	274	368	384	227	72	2	0
	CDD18.3	321	0	0	0	1	4	44	114	129	28	0	0	0
	CDH23.3	3233	0	0	0	19	120	436	1001	1294	355	8	0	0
	CDH26.7	1160	0	0	0	2	18	112	372	559	97	0	0	0
Wind	WSAvg	1.0	0.5	0.8	1.4	1.2	1.0	1.3	1.4	1.4	1.0	0.6	0.5	0.5
Precipitation	PrecAvg	596	31	28	39	55	81	79	63	52	46	49	41	30
	PrecMax	844	88	61	80	90	130	127	131	123	80	100	130	76
	PrecMin	438	0	6	13	25	43	17	6	18	12	8	0	4
	PrecStd	97	18	14	19	18	26	28	33	26	20	24	33	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	13.9	19.9	25.7	28.6	31.6	34.5	35.7	32.1	24.4	17.3	12.6
		MCWB	4.2	6.8	11.2	15.8	17.9	20.2	22.0	22.9	20.6	15.2	9.7	7.0
	2%	DB	7.4	11.4	16.8	22.6	26.2	29.4	32.5	33.7	29.3	22.2	14.8	9.5
		MCWB	3.0	5.5	9.2	13.5	16.4	19.6	21.5	21.9	19.2	14.1	8.8	4.8
	5%	DB	5.5	9.2	14.6	20.3	24.1	27.6	30.5	31.8	27.1	20.3	12.7	7.5
		MCWB	2.1	4.5	8.2	12.3	15.5	18.8	20.8	21.1	18.1	13.4	7.7	3.8
	10%	DB	3.9	7.1	12.6	17.9	21.9	25.7	28.6	29.8	25.1	18.3	10.6	5.7
		MCWB	1.3	3.2	7.1	11.0	14.5	17.9	20.0	20.2	17.0	12.4	6.8	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	7.2	11.9	17.0	18.9	21.2	23.0	24.1	21.7	16.0	10.8	7.7
		MCDB	8.1	12.2	18.1	24.8	26.2	30.0	32.8	33.8	29.5	22.9	15.5	12.4
	2%	WB	3.6	5.9	9.8	14.0	17.2	20.1	21.9	22.6	19.8	14.8	9.4	5.4
		MCDB	6.4	10.7	15.5	21.2	24.6	28.2	31.1	32.1	28.1	20.8	13.3	8.5
	5%	WB	2.5	4.8	8.6	12.7	15.9	19.2	21.1	21.6	18.5	13.8	8.3	4.2
		MCDB	4.9	8.6	13.9	19.5	23.0	26.5	29.5	30.2	26.0	19.1	11.6	6.7
	10%	WB	1.6	3.6	7.4	11.4	15.0	18.3	20.3	20.6	17.4	12.9	7.3	3.3
		MCDB	3.5	6.7	11.9	17.1	20.9	24.7	27.6	28.7	23.9	17.1	10.3	5.5
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.8	12.5	13.8	13.3	13.3	13.3	15.1	14.6	12.9	11.8	9.8
		MCDBR	10.6	13.4	16.8	18.8	18.2	16.3	16.7	18.3	18.2	17.0	15.2	12.0
		MCWBR	6.8	8.2	9.8	10.0	8.6	7.1	6.8	7.6	8.4	8.9	9.1	7.7
	5% WB	MCDBR	9.1	12.5	15.2	17.3	16.4	15.2	15.4	17.3	17.4	15.2	13.6	10.6
		MCWBR	6.0	7.6	9.0	9.3	8.0	6.8	6.5	7.4	8.3	7.9	8.2	6.9
Clear-Sky Solar Irradiance	taub	0.305	0.345	0.407	0.468	0.431	0.411	0.444	0.446	0.382	0.378	0.325	0.298	
	taud	2.200	2.079	1.986	1.950	2.149	2.274	2.203	2.192	2.358	2.288	2.361	2.289	
	Ebn at Noon	825	840	826	804	847	864	830	816	850	801	804	806	
	Edh at Noon	96	128	159	178	150	133	141	138	109	104	83	81	
All-Sky Solar Radiation	RadAvg	1.90	2.97	3.82	4.65	5.24	5.93	6.00	5.60	4.54	3.04	2.05	1.59	
	RadStd	0.20	0.34	0.42	0.45	0.49	0.41	0.51	0.42	0.41	0.30	0.27	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page